- A traceable evaluation structure that distinguishes algorithm experiments from functional, integration, acceptance, performance, reliability, and security evidence, while explicitly documenting threats to validity.

| Thesis objective | Implemented contribution | Evaluation evidence |
|---|---|---|
| Analyze IT recruitment workflows | Role-specific requirements and Human-in-the-Loop boundaries | Use cases and traceability matrix |
| Implement CV–JD matching | TF-IDF, cosine scoring, explanations, and persisted Matchings | Algorithm benchmark and backend tests |
| Learn from feedback | Rocchio updates followed by scheduled recomputation | Baseline-versus-learned ranking metrics |
| Support controlled automation | AutoFit policy, quota, cooldown, notification, and audit controls | Security tests and P0 E2E flows |
| Deliver an integrated platform | Spring Boot, React/TypeScript, PostgreSQL, and Flyway | Build, runtime health, and browser evidence |

*Table 1.1. Mapping of objectives, contributions, and evidence*

## 1.6 Thesis Organization

The thesis contains six chapters. Chapter 1 introduces the problem, objectives, scope, and contributions. Chapter 2 presents the theoretical background and related work, including TF-IDF, cosine similarity, Rocchio feedback, ranking metrics, Human-in-the-Loop control, and relevant security foundations.

Chapter 3 covers requirements, use cases, architecture, data design, security, and deployment. Chapter 4 explains the implementation. Chapter 5 presents the evaluation method and verified results, and Chapter 6 discusses achievements, limitations, future work, and the final conclusion.